# KUO, CHEN CHIH 郭晨志

M.S. in Chemical Engineering • Licensed environmental specialist

**Seeking** **EHS Engineer**

Email b29374665@gmail.com
Website b29374665-design.github.io
Xinzhuang, New Taipei · Open to relocation / shifts

## Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (**wastewater, waste, indoor air**), **OSH** exams scheduled; hands-on with **hazardous materials** (lithium metal, organic-solvent electrolytes, faulty Li–$SOCl_2$ cells), lab safety and **thermal hazard analysis (TGA / DSC)**.

## Experience

**2025 – Present** — **Sustainable Electrochemical Energy Development (SEED) Center, NTUST**
Lab Equipment Procurement & Asset Maintenance Manager

- **Procurement (>NTD 2M):** Led procurement of glovebox $O_2$ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes, low-temperature baths and TGA platinum pans; also replace instrument consumables.
- **Lab safety:** monitor glovebox $O_2$ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.
- Assembled cells and prepared electrodes in a battery dry room, working in cleanroom garments under clean-area gowning and entry procedures.
- **Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li–$SOCl_2$ cells.
- **TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's.
- **Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.

**Summer 2022** — **New Taipei City Dept. of Urban Development**
Summer Intern, Housing Development Section

- Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.

**2021 – 2024** — **Tunghai University Student Housing / GEPT**
Resident Assistant (2 yrs) / Exam Staff (part-time)

- Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

## Research & Industry

### Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

**Screening methodology**
Designed full-stage test protocols, experimental workflows and statistical validation models.

**Advanced analysis**
EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

**Database & SOP**
Established the test database and standardized high-throughput screening procedures (SOP).

**Training & tech transfer**
Trained QA and R&D teams to independently run large-batch screening and data interpretation.

**Nature Energy** Journal submission — battery thermal stability study Status **Under review**
**Co-author** Responsible for TGA thermal measurement
Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

## Conferences & Training

**Presenter** **2026 TwIChE Annual Meeting**
2026/12/05–06 • Feng Chia University
Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li ‖ SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability*

**Vendor seminar** **TA Instruments ARES-G3 launch & applications forum**
2026/07/17 • National Taiwan University

**Vendor lecture** **TA Instruments TGA lecture**
2026/06/03 • National Taiwan University

## EHS Licenses

- Dedicated Wastewater and Sewage Treatment Specialist (Class B) — Certified 2026
- Indoor Air Quality Maintenance Management Specialist — Certified 2026
- Waste Disposal Technician (Class A) — Certified 2026
- Occupational Safety and Health Management Personnel (Level B) — 2026 / 11 · mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A) — 2027 / 03 · exam scheduled

## Education

**2025 – 2027 (expected)**
**National Taiwan University of Science and Technology**
M.S. in Chemical Engineering

Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.

**2020 – 2024**
**Tunghai University**
B.S. in Chemical & Materials Engineering

Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

## Skills & Instruments

**EHS fundamentals**
Chemical labelling rules (GHS) · Explosive limits (LEL / UEL) · Occupational exposure limits · Ventilation

**Electrochemistry & batteries**
Impedance spectroscopy (EIS) · Battery aging analysis · Reliability analysis · Low-temp electrolytes

**Process & operations**
Lithography (nanoimprint / self-assembly) · Equipment procurement & maintenance

**Software**
Aspen Plus V14 · Aspen Dynamics · Microsoft Office · OriginPro · TA Universal Analysis · TA TRIOS

## Instruments

**Maintenance**
TGA Q500 · DSC 2500

**Operation**
ARES-G3 · Anton Paar DMA 5002 · Glovebox

**Procurement**
Glovebox $O_2$ sensor · DSC sample press · Hydraulic electrode punch · Vacuum pump · Eppendorf Pipette · Low-temperature bath